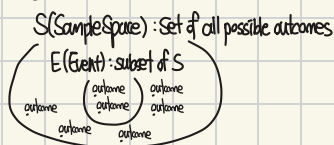


* Basics

Experiment

Initial settings $\rightarrow$ perform $\rightarrow$ get outcome



"Probability"

* Def: Bayesian def: Our belief that an event E will occur, expressed in number between 0 and 1
 Frequentist def: $P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$ $\rightarrow$ E must be repeated many times

* Axioms $\begin{cases} 0 \leq P(E) \leq 1 \\ P(S) = 1 \\ \text{If } E \cap F = \emptyset \text{ (mutually exclusive/disjoint), then } P(E) + P(F) = P(E \cup F) \end{cases}$

* Corollary $\begin{cases} P(E^c) = 1 - P(E) \quad (P(S) - P(E)) \\ E \subseteq F, \text{ then } P(E) \leq P(F) \\ P(E \cup F) = P(E) + P(F) - P(E \cap F) \end{cases}$
 General inclusion-exclusion principle

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n (-1)^{i+1} \sum_{i_1 < \dots < i_i} P(E_{i_1} E_{i_2} \dots E_{i_i})$$

* Equally likely outcomes: Define S as a sample space with equally likely outcomes, then $P(E) = \frac{|E|}{|S|}$

* Conditional probability: $P(A|B) := \frac{P(A \cap B)}{P(B)}$

$$\text{Bayes rule: } P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

$$\text{Conditioned bayes rule: } P(B|A \cap C) = \frac{P(B|C)P(A|B \cap C)}{P(A|C)}$$

* Chain rule

For any n events $A_1, \dots, A_n$ the joint probability can be expressed as a product of conditionals

$$\begin{aligned} P(A_1 \cap A_2 \dots A_n) \\ = P(A_1) P(A_2|A_1) P(A_3|A_2 \cap A_1) \dots P(A_n|A_{n-1} \cap A_{n-2} \cap \dots \cap A_1) \end{aligned}$$

* Law of total probability

Let $B_1, \dots, B_n$ be n disjoint events whose union is the entire sample space.

$$\text{Then for any event } A, P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A|B_i) P(B_i)$$

We can then rewrite the bayes rule as follows:

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{P(A)} = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

* Independence

Events A, B are independent ($A \perp B$) if $P(A \cap B) = P(A)P(B)$

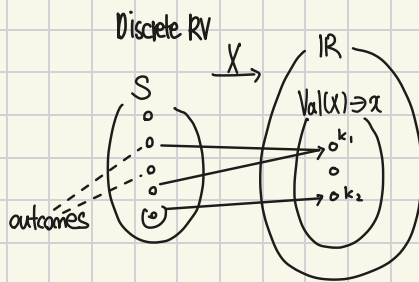
$$\text{So if } A \perp B, P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

In general, we call events $A_1, \dots, A_n$ are mutually independent if

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i) \text{ for any subset } S \subseteq \{1, \dots, n\}.$$

Random Variable: A variable (function) that encodes outcomes to numbers.

$X=k \Leftrightarrow$ event that random variable X takes on value k



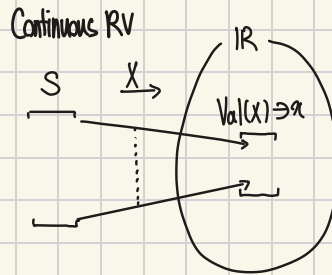
* Functions that map ω to probabilities

• PMF (Probability mass function)

$$p_X(x) := p(x) := P(X=x)$$

$P(X=k)$ can be nonzero

For a valid PMF, $\sum_{x \in \text{Val}(X)} p_X(x) = 1$.



* Functions that map ω to probabilities

• CDF (Cumulative distribution function) $\rightarrow$ PDF (Probability density function)

$$F_X(a) := F(a) := P(X \leq a)$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

$$\text{If } a \leq b \text{ then } F_X(a) \leq F_X(b)$$

$$P(a \leq X \leq b) = F_X(b) - F_X(a)$$

derivative of CDF

$$f_X(x) := f(x) := \frac{dF_X(x)}{dx}$$

$$P(a \leq X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x) dx$$

$$\text{for all real numbers } x, f_X(x) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$p_X(k) = 0$ for all k . $P(a \leq X \leq b)$ can be nonzero.

* Joint distribution between many RVs

Discrete RVs (Given $X_1, \dots, X_n$)

Joint PMF

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

$$\Leftrightarrow \sum_{x_1} \sum_{x_2} \dots \sum_{x_n} p(x_1, \dots, x_n) = 1$$

Marginal PMF (X_i)

$$p_{X_i}(x_i) = \sum_{x_1 \in \text{Val}(X_1)} \dots \sum_{x_n \in \text{Val}(X_n)} p(x_1, \dots, x_n)$$

$$E[g(X_1, X_2, \dots, X_n)] = \sum_{x_1} \dots \sum_{x_n} g(x_1, \dots, x_n) \cdot p(x_1, \dots, x_n)$$

Continuous RVs (Given $X_1, \dots, X_n$)

Joint PDF

$$f(x_1, \dots, x_n) = \frac{\partial^n F(x_1, \dots, x_n)}{\partial x_1 \partial x_2 \dots \partial x_n}$$

$$\Leftrightarrow \int_{x_1} \int_{x_2} \dots \int_{x_n} f(x_1, \dots, x_n) dx_1 \dots dx_n = 1$$

Marginal PDF (X_i)

$$f_{X_i}(x_i) = \int_{x_2} \dots \int_{x_n} f(x_1, \dots, x_n) dx_2 \dots dx_n$$

$$E[g(X_1, \dots, X_n)] = \int_{x_1} \int_{x_2} \dots \int_{x_n} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \dots dx_n$$

* Playing a bit more with distributions

- Conditional distribution

$$\text{DRV: } p_{Y|X}(y|x) = \frac{p_{XY}(x,y)}{p_X(x)}$$

$$\text{CRV: } f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$\text{generalize: } f_{X_1 | X_2, \dots, X_n}(x_1 | x_2, \dots, x_n) = \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{X_2, \dots, X_n}(x_2, \dots, x_n)}$$

- Bayes Rule

$$\text{DRV: } p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y) p_Y(y)}{\sum_{y \in \text{Val}(Y)} p_{X|Y}(x|y) p_Y(y)}$$

$$\text{CRV: } f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y) f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x|y') f_Y(y') dy'}$$

- Chain Rule

$$f(x_1, \dots, x_n) = f(x_1) f(x_2 | x_1) \dots f(x_n | x_1, \dots, x_{n-1})$$

$$= f(x_1) \prod_{i=2}^n f(x_i | x_1, \dots, x_{i-1})$$

- Independence

$$X \perp Y \Leftrightarrow F_{XY}(x,y) = F_X(x) F_Y(y) \text{ for all values of } x,y$$

$$\text{so for independent RVs } X_1, \dots, X_n: f(x_1, \dots, x_n) = f(x_1) f(x_2) \dots f(x_n) = \prod_{i=1}^n f(x_i)$$

* Expectation (probability-weighted average of values)

Let g be an arbitrary real-valued function.

$$(DRV) E[g(X)] := \sum_{x \in \text{Val}(X)} g(x) p_X(x)$$

$$(CRV) E[g(X)] := \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

• for any constant $a \in \mathbb{R}$, $E[a] = a$ and $E[ag(X)] = aE[g(X)]$

• Independent RVs $\Rightarrow E(XY) = E(X)E(Y)$

• Conditional Expectation

$$E(X|Y) = \sum_{x \in \text{Val}(X)} x p_{X|Y}(x|y) \rightarrow \text{function of } Y$$

• Linearity of expectation

$$\text{Given } n \text{ real-valued functions } g_1(x) \dots g_n(x), E\left[\sum_{i=1}^n g_i(x)\right] = \sum_{i=1}^n E[g_i(x)]$$

• Law of total expectation

$$\begin{aligned} \text{Given two RVs } X, Y: E[E(X|Y)] &= \sum_{y \in \text{Val}(Y)} E(X|Y=y) p_Y(y) = \sum_{y \in \text{Val}(Y)} p_Y(y) \cdot \sum_{x \in \text{Val}(X)} x \cdot p_{X|Y}(x|y) \\ &= \sum_{y \in \text{Val}(Y)} \sum_{x \in \text{Val}(X)} x \cdot p_{X,Y}(x,y) = \sum_{x \in \text{Val}(X)} x \sum_{y \in \text{Val}(Y)} p_{X,Y}(x,y) = \sum_{x \in \text{Val}(X)} x p_X(x) = E(X) \end{aligned}$$

* Covariance

Measures how much one RV's value tends to move w another RV's value

$$\text{Cov}[X, Y] := E[(X - E(X))(Y - E(Y))] = E[XY] - E[X]E[Y]$$

> 0 : positively correlated

< 0 : negatively correlated

$= 0$: uncorrelated

• If $X \perp Y$, then $E[XY] = E[X]E[Y]$, so $\text{Cov}[X, Y] = 0$.

• $\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]$ / $\text{Var}[X_1 + \dots + X_n] = \text{Var}[X_1] + \dots + \text{Var}[X_n] + 2 \sum_{i < j} \text{Cov}[X_i, X_j]$

• $\text{Cov}[X, X] = E[XX] - E[X]E[X] = \text{Var}[X]$

* Random Vector

$$\text{Random Vector } X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} \text{ RVs}$$

All the joint PDF/CDF notion will apply to X .

$$\text{Given } g: \mathbb{R}^n \rightarrow \mathbb{R}^m, g(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix}, E[g(X)] = \begin{bmatrix} E[g_1(X)] \\ \vdots \\ E[g_m(X)] \end{bmatrix}$$

* Covariance Matrix

For a random vector $X \in \mathbb{R}^n$, we define its covariance matrix Σ

$$\text{as: } \Sigma = \begin{bmatrix} \text{Cov}[X_1, X_1] & \dots & \text{Cov}[X_1, X_n] \\ \vdots & \ddots & \vdots \\ \text{Cov}[X_n, X_1] & \dots & \text{Cov}[X_n, X_n] \end{bmatrix}$$

$$= E[(X - E(X))(X - E(X))^T]$$

• symmetric and PSD

$\Rightarrow$ • if $X_i \perp X_j$ for all i, j , then $\Sigma = \text{diag}(\text{Var}[X_1], \dots, \text{Var}[X_n])$

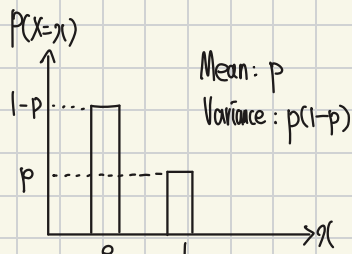
* Some special probability distributions (1 RV)

Discrete RV

Bernoulli(p)

- One yes/no trial (success rate p)

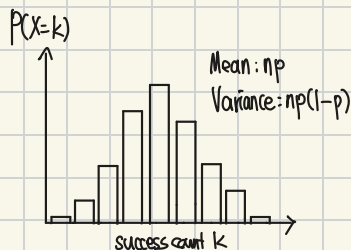
$$p_X(x) = p^x (1-p)^{1-x} \quad (x \in \{0, 1\})$$



Binomial(n, p)

- number of successes in n independent yes/no trials (success rate p)

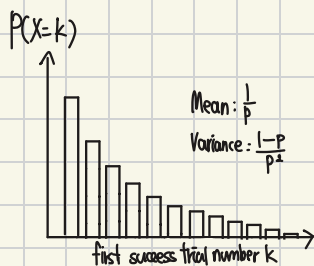
$$p_X(k) = n C_k \cdot p^k (1-p)^{n-k} \quad (k = 0, 1, \dots, n)$$



Geometric(p)

- number of independent yes/no trials needed to get the first success (success rate p)

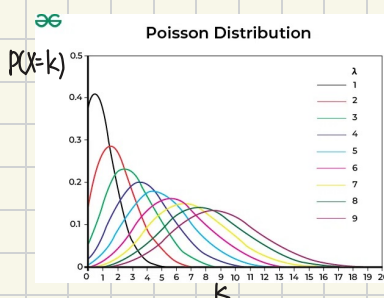
$$p_X(k) = p(1-p)^{k-1} \quad (k = 1, 2, \dots)$$



Poisson(λ)

- number of times a rare event occurs in a fixed interval of time and space, given an average rate of λ occurrences

$$p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad (k = 0, 1, \dots)$$



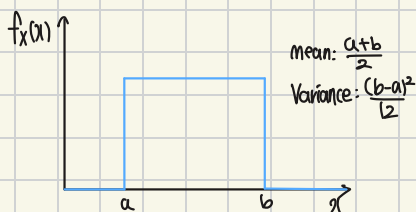
Mean: λ
Variance: λ

Continuous RV

Uniform(a, b)

every value in the interval (a, b) is equally likely, with constant probability density and zero density outside the interval

$$f_X(x) = \frac{1}{b-a}, \quad (x \in (a, b))$$



Gaussian(μ, σ²)

bell-shaped distribution symmetric around its mean μ, with σ² controlling how spread out they are

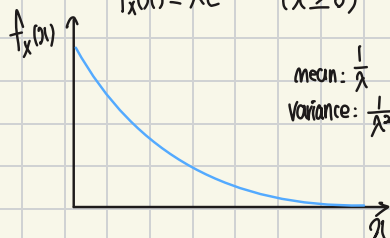
$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (x \in (-\infty, \infty))$$



Exponential(λ)

continuous waiting time until an event occurs given a constant occurrence rate λ (continuous counterpart of geometric)

$$f_X(x) = \lambda e^{-\lambda x} \quad (x \geq 0)$$



* Multivariate Gaussian ($X \sim N(\mu, \Sigma)$)

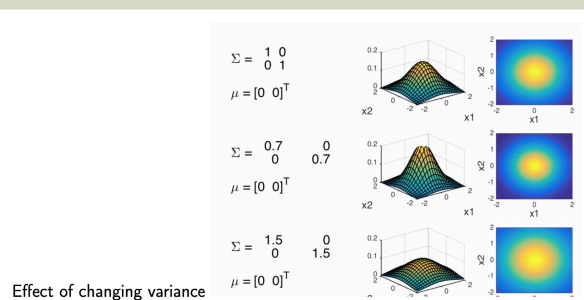
$$X = \text{Random Vector} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \mu = \text{mean vector } E[X] = \begin{bmatrix} E[x_1] \\ \vdots \\ E[x_n] \end{bmatrix} \quad \Sigma = \text{Covariance matrix of } X$$

$$\text{PDF: } f(x; \mu, \Sigma) = \frac{1}{\det(\Sigma)^{\frac{1}{2}} (2\pi)^{\frac{n}{2}}} \cdot e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

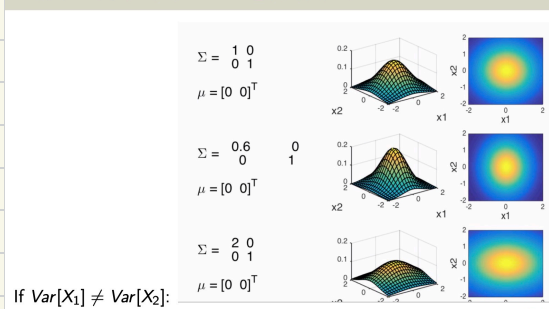
Some nice properties of MV Gaussian

- Marginals and conditionals of a joint Gaussian are Gaussian
- A d -dimensional Gaussian $X \in N(\mu, \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2))$ is equivalent to a collection of d independent Gaussians $X_i \in N(\mu_i, \sigma_i^2)$.
This results in isocontours aligned with the coordinate axes.
- In general, the isocontours of a MV Gaussian are n -dimensional ellipsoids with principal axes in the directions of the eigenvectors of covariance matrix Σ (Σ is PSD so all n eigenvectors are non-negative)
The axes' relative lengths depend on the eigenvalue of Σ .

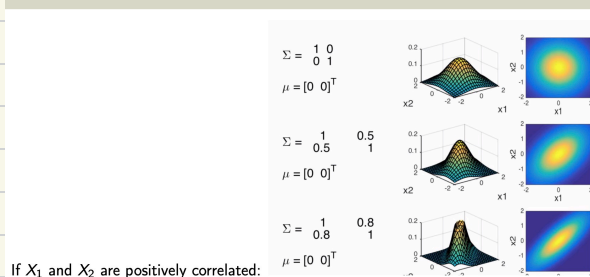
Visualizations of MV Gaussians



Visualizations of MV Gaussians



Visualizations of MV Gaussians



Visualizations of MV Gaussians

